



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO K. G. FUEL STATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

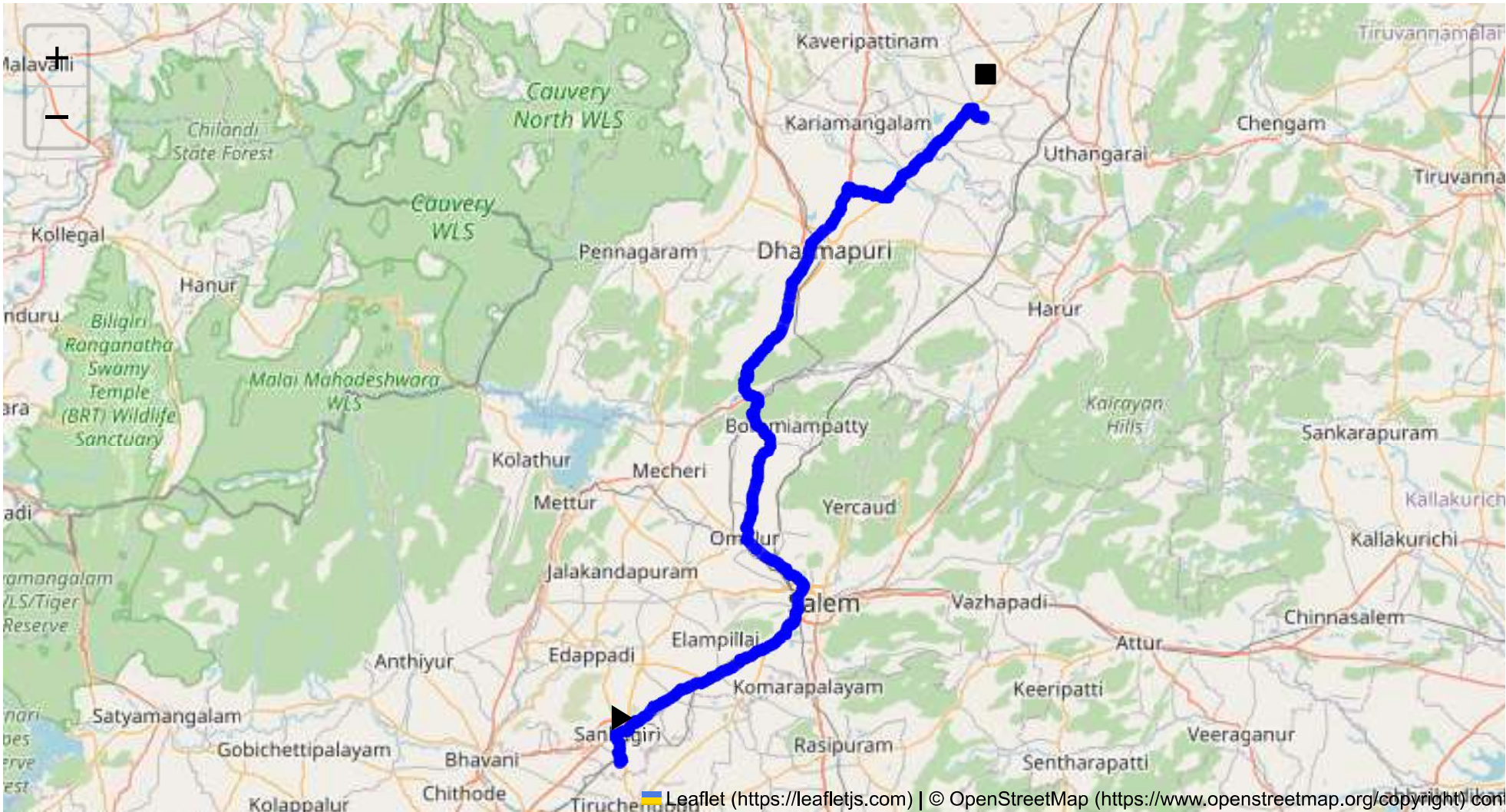
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 144.31 km
Estimated Duration: 2.9 hours
Adjusted Duration (Heavy Vehicle): 3.6 hours
Start: (11.4381, 77.8734)
End: (12.3137925, 78.3799343)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri (CVQF+23W) to Ayalampatti, Pochampalli (897H+GRM) covers approximately 144.31 kilometers. The journey typically entails traveling along National Highways and regional roads, moving through both urban and rural environments. Key towns along the way may include major junctions such as Salem and Dharmapuri, which are likely hubs for traffic and commerce.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate. During summer (March-June), temperatures exceed 40°C, which can lead to road surface degradation. Monsoon season (October-December) can see heavy rains, leading to potential flooding and reduced visibility. It's important to check weather forecasts before the journey, as sudden rains can cause slippery roads and impact driving conditions.

3. Analysis of Traffic Patterns

- **Peak Hours:** Typically, traffic congestion can be significant during early morning hours (7-9 AM) and late evening (5-8 PM), especially near urban centers like Salem and Dharmapuri.
- **Congestion Areas:** Expect heavier traffic near city centers, toll booths, and market areas. National highways see high truck volume, which may slow transit during peak hours.

4. Assessment of Road Quality and Infrastructure

- **Highways:** Generally, National Highways are well-maintained with good signage. However, occasional potholes and uneven surfaces may be encountered.
- **Regional Roads:** These may have varying quality, including narrow lanes and limited shoulder space. Be cautious of sharp turns and rural traffic (including livestock).

5. Suggestions for Alternative Routes for Emergencies

- **Alternative 1:** If the primary National Highway is blocked, consider smaller state highways that parallel the main route.
- **Alternative 2:** Routes through less congested rural roads might provide a detour around heavily trafficked areas or accidents but at slower speeds.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu state laws require permits for transporting hazardous materials and often designate specific hours for transport to minimize public risk. Ensure compliance with all documentation, proper vehicle markings, and follow restricted area guidelines.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

Historically, there have been incidents of hazardous material spills and vehicle collisions, particularly on hilly or curving sections. Pay special attention to accident-prone areas near city ingress points and during adverse weather conditions.

8. Environmental Considerations and Sensitive Areas

Some segments of the route may run close to environmentally sensitive areas, including water bodies and agricultural lands. Be mindful of emissions and potential spillage, adhering strictly to environmental guidelines during transit.

9. Analysis of Communication Coverage

Mobile network coverage is generally reliable along National Highways in Tamil Nadu. However, expect potential dead zones in rural areas, particularly in valley areas or dense forests around hill sections.

10. Estimated Emergency Response Times for Different Route Segments

- Urban Sections:** Emergency response (police, fire, medical) typically within 30-45 minutes.
- Rural Areas:** Response times may extend to 60-90 minutes due to distance from major towns and limited infrastructure.

11. An Overall Summary of Risk Assessment

- Weather Risks:** High during monsoon; potential for heat-related road issues in summer.
- Traffic Risks:** Congestion likely in urban areas; higher accident exposure near city junctions.
- Infrastructure Risks:** Generally sound on National Highways; varies on regional roads.
- Regulatory Risks:** Strict adherence to hazardous material transport regulations is crucial.
- Communication and Emergency Risks:** Adequate in urban but weaker in rural segments.

Overall, the journey from Sangagiri to Ayalampatti involves moderate risk due to traffic, road conditions, and weather-related issues. Preparation in terms of alternative routes, regulatory compliance, and communication planning is essential for safe transit.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45028, 77.87202	15 KM/Hr
4	Turn	Medium	11.46400, 77.87323	30 KM/Hr
5	Turn	High	11.47403, 77.86841	15 KM/Hr
6	Turn	High	11.49149, 77.89601	15 KM/Hr
7	Turn	High	11.49224, 77.89601	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
8	Turn	Medium	11.53512, 77.95717	30 KM/Hr
9	Turn	Medium	11.53606, 77.95869	30 KM/Hr
10	Turn	Medium	11.62403, 78.10893	30 KM/Hr
11	Turn	High	11.64183, 78.11810	15 KM/Hr
12	Turn	Medium	11.64338, 78.11998	30 KM/Hr
13	Turn	Medium	11.67903, 78.12796	30 KM/Hr
14	Turn	Medium	11.73968, 78.05098	30 KM/Hr
15	Turn	Medium	11.86790, 78.08347	30 KM/Hr
16	Turn	Medium	11.91357, 78.05868	30 KM/Hr
17	Turn	Medium	11.92556, 78.06499	30 KM/Hr
18	Turn	Medium	11.93694, 78.05425	30 KM/Hr
19	Turn	Medium	11.95531, 78.04698	30 KM/Hr
20	Turn	High	11.95803, 78.04964	15 KM/Hr
21	Turn	Medium	11.96068, 78.04960	30 KM/Hr
22	Turn	High	11.96370, 78.05142	15 KM/Hr
23	Turn	High	11.96542, 78.04938	15 KM/Hr
24	Turn	High	11.96650, 78.04927	15 KM/Hr
25	Turn	High	12.21900, 78.19171	15 KM/Hr
26	Turn	Medium	12.21619, 78.19634	30 KM/Hr
27	Turn	Medium	12.21373, 78.21093	30 KM/Hr
28	Turn	High	12.21454, 78.21229	15 KM/Hr
29	Turn	High	12.21200, 78.21640	15 KM/Hr
30	Turn	Medium	12.21232, 78.21749	30 KM/Hr
31	Turn	High	12.21131, 78.22316	15 KM/Hr
32	Turn	Medium	12.21082, 78.22342	30 KM/Hr
33	Turn	Medium	12.20520, 78.24481	30 KM/Hr
34	Turn	Medium	12.24544, 78.28194	30 KM/Hr
35	Turn	Medium	12.26519, 78.30309	30 KM/Hr
36	Turn	Medium	12.26745, 78.30399	30 KM/Hr
37	Turn	Medium	12.27014, 78.30741	30 KM/Hr
38	Turn	Medium	12.28390, 78.31963	30 KM/Hr
39	Turn	Medium	12.28461, 78.32199	30 KM/Hr
40	Turn	High	12.32698, 78.36027	15 KM/Hr
41	Turn	Medium	12.32663, 78.36196	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
42	Turn	High	12.32602, 78.36486	15 KM/Hr
43	Turn	High	12.32118, 78.36543	15 KM/Hr
44	Turn	High	12.31899, 78.36814	15 KM/Hr
45	Turn	Medium	12.31833, 78.36806	30 KM/Hr
46	Turn	Medium	12.31750, 78.36854	30 KM/Hr
47	Turn	Medium	12.31569, 78.37192	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
6	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
7	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
9	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
10	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
11	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
12	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
13	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
15	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
20	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
21	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
22	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
24	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
25	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
26	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
27	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
28	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
29	hospital	Global Medical Centre Jagirammapalayam	11.6894409, 78.1108847	30 km/h	Medium
30	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
34	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
35	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
37	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
39	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
40	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
41	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium